

SUMMARY

Senior Python Developer & Data Engineer with 5+ years of experience building scalable, cloud-native data platforms and AI-driven applications. Expert in **Python, FastAPI, Apache Spark**, and real-time **ML workflows**. Proven track record in designing microservices, leading small teams, optimizing ETL pipelines (reducing runtime by up to 60%), and delivering actionable insights through data-driven solutions.

EMPLOYMENT

Lead Python Developer Virtusa (Consulting for Citi Bank) Jun'24 – Present

- Designed and implemented microservices with REST APIs using **FastAPI** to extract, validate, transform, and load metadata into MongoDB and Oracle database, supporting input from XLSX file, and database sources.
- Migrated data validation scripts to Python, utilizing vectorized operations to reduce **execution time by 60%**, and deployed as a scalable microservice to support future data growth, optimizing request throughput.
- Lead developer for Maphub project, driving architecture design, implementation, and mentorship.
- Integrated **real-time ML training and inference** using **AWS AutoGluon**, enabling a fully UI-driven workflow where users can train, version, and deploy models for live predictions via the **Maphub Data Studio**.

Specialist programmer Infosys May'19 – Dec'21

- Engineered a scalable **streaming data pipeline** on GCP, utilizing Pub/Sub and Cloud Run for automated data ingestion and processing. This enabled real-time insights into store trends, driving a **15% increase in sales** and accelerating data-driven decision-making.
- Built and managed independently four complex **ETL batch data pipelines** using Apache PySpark, Delta Lake, Databricks, Apache Airflow, and AWS services, serving **three clients**: Levi's, Kraft Heinz, and HCSC.
- Optimized data processing performance and resource utilization by strategically implementing window functions, leading to **30% reduction in Spark job runtime** and 15% decrease in cluster memory usage.

EDUCATION

- Northeastern University** - Master's in computer science (Jan'22 – Dec'23)
 - Teaching Assistant** - Database Management Systems; Human-Computer Interaction
 - Research Assistant** - List Curator (Summer'22 - Fall'23)
- Amrita School of Engineering** - Bachelor of Technology in Computer Science (Aug'15 – May'19)

PROJECTS

- [Wander Finds](#) (Gemini AI Hackathon'24). AI-driven mobile app that generates personalized travel recommendations based on Google Maps data and user preference. **Flutter, Firebase, LangChain, Gemini**.
- [Shark Tank Analysis](#) (Fall'23). Developed an interactive dashboard to analyze entrepreneur participation across industries and states, highlighting trends in valuations. **Data Analysis, D3.js, JavaScript, Tableau**.
- [Hate Speech Detection](#) (Summer'22). Developed an in-browser hate speech detection dashboard, enabling real-time text classification as hate, offensive, or neutral categories. **ML, XGBoost + TF-IDF, Pyodide**.

SKILLS

- Languages**: Python; JavaScript; Java; Bash; C
- Databases**: MongoDB; MySQL; Postgres; DuckDB; SQLite
- Certifications**: [PCAP™ – Certified Associate Python Programmer](#); Smart Analytics, ML, and AI on GCP
- Workflow & Platforms**: Delta Lake; Apache Spark; Airflow; Databricks; Tableau; D3.js
- ML Tools**: AWS AutoGluon; GCP Cloud Run; LangChain; Scikit-learn; XGBoost; Pyodide

PUBLICATION

- Author**: Vandanapu kasi. Hadoop and Natural Language Processing Based Analysis on Kisan Call Center (KCC) Data; 2018 International Conference on Advances in Computing, Communication, and Informatics.